

Md Rafi Ahmed

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About Me

Aspiring Computer Science student with a strong foundation in C, JavaScript, and full-stack web development. Passionate about building functional, user-centric applications, as demonstrated through projects like autonomous voice-enabled web agents and full-stack healthcare platforms. Eager to leverage my problem-solving skills in DSA and modern web technologies to contribute to innovative software solutions and collaborative technical environments.

Skills

- **Languages:** C, JavaScript, HTML, CSS, DSA (Data Structures & Algorithms)
- **Core Concepts:** Object-Oriented Programming (OOP), Python
- **Tools:** Git, GitHub

Projects

Thanking Portal

- Developed a Thanking Portal that enables users to create, personalize, and share appreciation messages for friends through an intuitive and user-friendly interface.
- Designed to foster positive social connections by preserving meaningful messages and creating a lasting digital space for gratitude and memories.

Doctor Appointment Website

- Developed a full-stack Doctor Appointment Website with secure patient registration, authentication, and appointment booking functionality.
- Implemented features for appointment scheduling, doctor availability management, and patient appointment history with a responsive user interface.
- Built an end-to-end healthcare management platform with database integration, CRUD operations, and a seamless user experience.

C Inventory Management System

C, File I/O, Git

- Developed a lightweight, console-based inventory management application utilizing core C programming concepts.
- Implemented CRUD (Create, Read, Update, Delete) functionality using standard binary file I/O operations for data persistence.
- Engineered robust error handling, including NULL pointer checks, to prevent program crashes during missing database file scenarios.

Smart Automated Garage System

Arduino Uno, C++, Hardware Automation

- Built a fully automated, bidirectional smart gate system applying custom logic for seamless vehicle detection and entry control.
- Integrated and programmed an Arduino Uno with dual HC-SR04 ultrasonic sensors and an SG90 servo motor for precise mechanical actuation.
- Constructed a functional physical cardboard prototype to successfully demonstrate real-world hardware and software integration.

WebPilot | Autonomous Voice-Enabled Web Agent

- Architecting an autonomous, voice-activated web agent using FastAPI, Playwright, Gemini API, and HTML/CSS/JS.

- Developing the full-stack architecture, utilizing FastAPI for backend routing and Playwright for robust web automation and task execution.
- Integrating the Gemini API to process natural language voice commands and translate them into autonomous browser actions seamlessly.

Technical Practice & Implementation

Programming Fundamentals (C, DSA)

- Have a strong grip on basic console programs using conditional statements, loops, functions, and arrays in C.
- Practiced problem-solving using recursion, sorting algorithms, and user-defined functions.
- Implemented core data structures like arrays and basic linked lists.

Beginner Python Practice

- Built simple console applications and scripts to grasp Python syntax, loops, and logic.
- Performed file input/output operations for reading, parsing, and writing data efficiently.

Education

B.Tech in Computer Science and Engineering (2025-2029)

Pursuing

JIS College of Engineering, Kalyani

Interest

- Participating in hackathons and beginner-friendly tech events
- Web Designing
- Exploring core computer science concepts